🚀 CityConnect – Smart Civic Engagement Platform

A Smart City Web Platform to Bridge Citizens & Government for Cleaner, Safer, and Smarter Cities.

🧠 What We're Building (In Simple Terms)

CityConnect is a **smart city web platform** that allows:

👥 **Citizens** to:

- Report civic issues (potholes, garbage, broken lights, etc.)
- Earn EcoCoins by contributing to green activities like e-waste donation or cleanup tasks
- Redeem those coins for rewards like discounts, gifts, coupons
- Track their reports, see public issues in their area
- Stay updated via News or Alerts posted by government admins
- Get support through an AI Chatbot or WhatsApp bot

🏛️ **City Departments/Admins** to:

- Receive only the complaints relevant to their department (e.g. PWD, Waste Mgmt)
- Resolve and update the status of issues
- Approve EcoCoin reward redemptions
- Post city-level news or alerts
- Manage performance and issue load via dashboards

🎯 What Is the Goal of This Platform?

To **digitally connect citizens with real municipal departments** in a professional, gamified way — enabling transparency, cleaner cities, and civic engagement.
It also rewards good behavior (recycling, donating, reporting) through **EcoCoins**.

🧱 What We're Building – Full Platform Scope

👥 User Side (Public Site) – 14 Pages

Page	Purpose
Home	Project intro, impact, CTA
Report Issue	Form to submit complaints with photo + location
My Reports	Track status of your complaints

Tasks	Do civic tasks (donation, cleanup), earn coins
Feed	See public issues reported by others
Store	Redeem EcoCoins for real rewards
Redeem History	Track past redemption requests
Leaderboard	See top contributors in the city
Profile	See coins earned, tasks done, reports made
News	Alerts or updates from departments
Help	AI chatbot or static FAQs
Contact	Feedback or support
Login / Signup	Firebase Auth (Google login)
AI Assistant	Chatbot to help with platform usage (floating)

👩‍💼 Admin Panel – 8 Role-Based Dashboards

Each admin logs in with their department and sees **only their relevant data**.

Role	Dashboard Features
Super Admin	See everything, manage admins, assign roles
Waste Management	Handle garbage issues
Public Works (PWD)	Handle road, lighting, infra issues
Recycling	Approve e-waste or donations, assign coins
Rewards & EcoCoins	Approve reward redemptions
Grievance Cell	Handle unresolved or escalated issues
News	Post public announcements, alerts
Admin Login	Role-based login with redirection

⚙️ How the System Works (Flow Summary)

1. 👤 **User logs in** and reports an issue (form with image + category + location)
2. 📩 **The system auto-routes** the issue to the correct admin panel (e.g. pothole → PWD)
3. 👩‍💼 **Admin logs in**, sees assigned issues, updates status to "Resolved" or "Rejected"
4. 🏆 If a user does a task (like donating e-waste), an admin verifies it → user earns **EcoCoins**
5. 🎁 User redeems coins in Store → request goes to Rewards Admin → they approve

6. 🌍 Admins post news/alerts → visible on the News page
7. 📲 Users can also use **WhatsApp** to report simple issues or ask for help

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👩‍💻 Tech Stack We're Using

Layer	Tool
Frontend	**React.js (Next.js App Router)**
UI/Styling	**Tailwind CSS** + **ShadCN UI** + **Inter/Poppins Fonts**
Backend	**Firebase** (Firestore, Auth, Storage)
Hosting	**Vercel** (frontend) + Firebase if needed
File Uploads	Firebase Storage
Auth	Google Sign-In
Chatbot	Static or OpenAI API (optional)
WhatsApp Integration	Meta Cloud API (basic demo or real)

🔥 What Makes Our Project Strong?

- **Real civic problem** + working system
- **Modular admin structure** just like real city offices
- **Gamified eco rewards** (EcoCoins)
- **Clean, professional UI** (not a "student" project)
- **Inclusive** via WhatsApp + web
- **Scalable & easy to deploy**

🛠️ Summary

We're building a full-stack Smart City platform with real reporting, real department routing, eco reward system, and a polished UI — like a real civic SaaS product.